Lecture-13
Conservation Laws for Control Volumes I: Introduction to Control Volumes and Reynolds Transport Theorem

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Learning Objectives

By the end of this lecture, you should be able to

1. Differentiate between a **system approach** and a **control volume approach** to a problem
2. Define an **extensive property** and its corresponding **intensive property**.
3. Explain the meaning of each term in the **Reynolds Transport Equation**.
4. Formulate the quantitative relationship between a **system's rate of change** and a **control volume's surface and volume integrals**.

System versus Control Volume Approach

- When we study the **motion of an individual particle or a group of particles** as they traverse through space, the approach is called the systems approach**.**
- This is the approach when we are using a **Lagrangian description** of motion**.**
- The control volume approach involves **studying a region in space** as a system of particles passes through it**.**
- The control volume approach is motivated by the **Eulerian description** of motion**.**

Intensive and Extensive properties

- Physical and chemical properties of materials can often be categorized into **extensive and intensive** properties.
- An extensive property is a physical quantity whose value is proportional to the size of the system it describes, or the **quantity of matter in the system**.
- An intensive property is a physical quantity whose value **does not depend on the amount of substance that was measured**.

- amount of substance, n
- enthalpy, H
- entropy, S
- Gibbs energy, G
- heat capacity, C_p
- Helmholtz energy, A or F
- internal energy, U
- spring stiffness, K
- mass, m
- volume, V

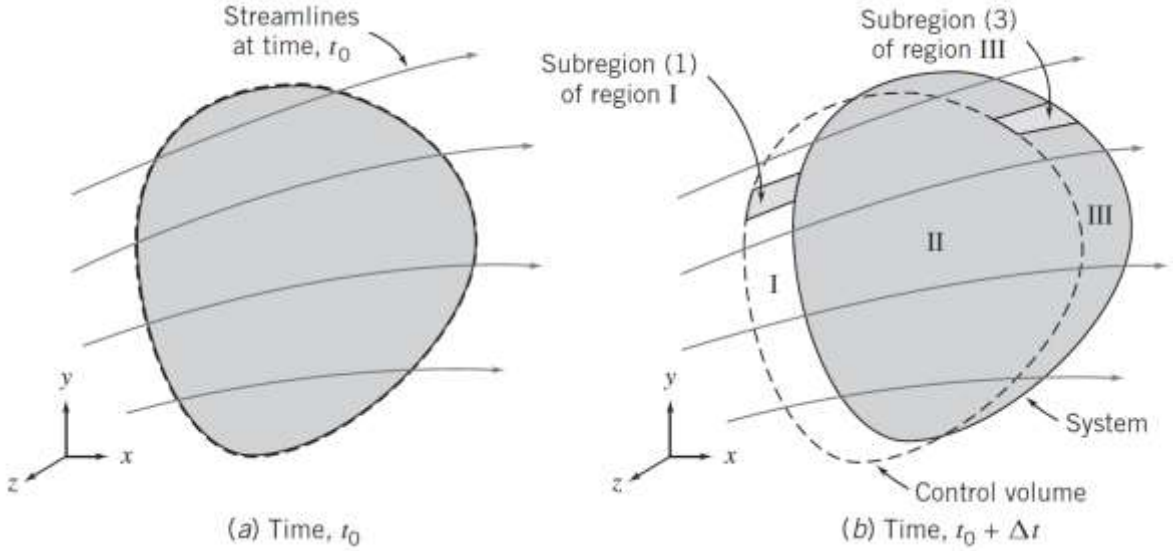
- charge density, ρ (or ne)
- chemical potential, μ
- color[6]
- concentration, c
- electrical conductivity (or specific conductance), σ (sometimes κ or γ)
- electric permittivity, ε
- electrical resistivity, ρ
- energy density, ρ
- magnetic permeability, μ
- mass density, ρ (or specific gravity)
- melting point and boiling point[7]
- molality, m or b
- molar mass, M

- molar volume, V_{m}
- pressure, p
- refractive index
- specific heat capacity, c_p
- specific internal energy, u
- specific rotation, $[\alpha]$
- specific volume, v
- standard reduction potential,[7] E°
- surface tension
- temperature, T
- thermal conductivity
- velocity v
- viscosity

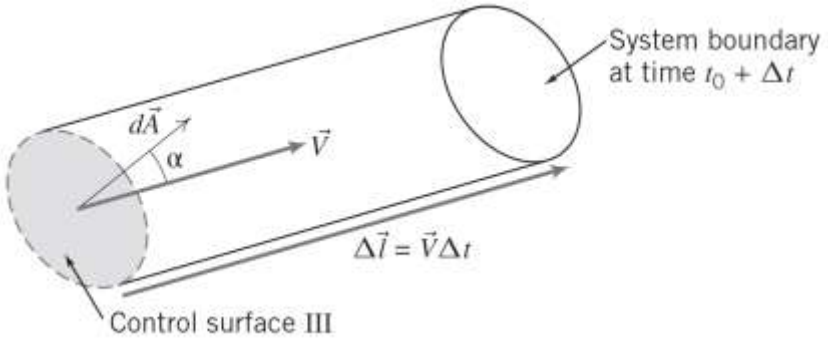
https://en.wikipedia.org/wiki/Intensive_and_extensive_properties

Reynolds Transport Theorem

$$N_{\text{system}} = \int_{M(\text{system})} \eta \, dm = \int_{\forall(\text{system})} \eta \, \rho \, d\forall$$



$$\left.\frac{dN}{dt}\right)_{\text{system}} \equiv \lim_{\Delta t \to 0} \frac{N_s)_{t_0+\Delta t} - N_s)_{t_0}}{\Delta t}$$



$$\left.\frac{dN}{dt}\right)_{\text{system}} = \frac{\partial}{\partial t}\int_{\text{CV}} \eta \, \rho \, d\forall + \int_{\text{CS}} \eta \, \rho \vec{V} \cdot d\vec{A}$$

- The **rate of change of any arbitrary extensive property of a system** can be related to the **variations of this property associated with a control volume** using the RTT.

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

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